# iLearn Practice Exam

12BR · Chapter / Topic 13

## Cambridge A Level Computer Science 9618

Paper 3  Advanced Theory

# Topic 13: Data Representation

**Name:** _____

**Class:** _____

**Date:** _____

**Time allowed:** 1 hour 30 minutes

**TOTAL MARKS: 72**

Marks obtained: ..... / 72

**INSTRUCTIONS**

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

**INFORMATION**

· The total mark for this paper is 72.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 12 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A programmer is writing a program to manage bookings for a small taxi company. The programmer requires some user-defined data types.

**(a)** Write a **pseudocode** statement to declare the enumerated data type, `Vehicle`, to hold the identity code of each of the company's taxis:

M100, M230, T101, T102, T120, T150

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..... [2]

**(b)** Write **pseudocode** statements to declare the composite data type, `Booking`, to hold data about taxi bookings. The data required includes:

- booking number (any combination of letters and numbers)
- destination
- client name
- client telephone number
- date of departure
- address for pick-up
- the identity code of the taxi used.

Use the most appropriate data type in each case, including the enumerated data type from part **(a)**.

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..... [4]

2 Numbers are stored in a computer using binary floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

**(a)** Write the normalised floating-point representation of the following binary number using this system.

0.00000011010111

**Mantissa** **Exponent**

[2]

**(b)** Calculate the normalised binary floating-point representation of –25.3125 in this system. Show your working.

**Mantissa** **Exponent**

Working .....

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[4]

3 **(a)** User-defined data types may be created by programmers when writing software.

Explain the reasons why user-defined data types are needed.

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**(b)** Write the pseudocode statement to declare a non-composite data type, `Fibonacci`, that holds the following numbers:

0, 1, 2, 3, 5, 8, 13, 21

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4 **(a)** State **one** example of where serial file organisation could be used.

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..... [1]

**(b)** Describe how records are organised in serial files.

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..... [2]

**(c)** Outline the process that is used to find a record in a serial file.

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..... [3]

5 Numbers are stored in a computer system using binary floating-point representation with:

- 12 bits for the mantissa
- 4 bits for the exponent
- two's complement form for both the mantissa and the exponent.

**(a)** Calculate the normalised binary floating-point representation of –18.5625 in this system.

Show your working.

**mantissa** **exponent**

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working .....

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[4]

**(b)** Explain the problem that would occur if the two binary floating-point numbers representing the denary numbers 91.5 and 123.25 were added together and the result stored in the given system.

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6 **(a)** State **one** example of where sequential file organisation could be used.

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..... [1]

**(b)** Describe how records are organised in sequential files.

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**(c)** Outline the process of sequential file access to find a record in a sequential file.

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..... [3]

7 Data types can be defined using pseudocode.

The composite record data type, `Departure`, is used to represent flights from Cambridge Airport and is defined in pseudocode as:

```
TYPE Departure
   DECLARE FlightNumber : STRING
   DECLARE Destination : STRING
   DECLARE FlightDate : DATE
   DECLARE Gate : STRING
   DECLARE Airline : STRING
ENDTYPE
```

A variable, `Flight1`, is declared in pseudocode as:

```
DECLARE Flight1 : Departure
```

**(a)** Write **pseudocode** to store the following details to `Flight1`:

| Field | Data |
|---|---|
| FlightNumber | SB2789 |
| Destination | Dublin |
| FlightDate | 30/07/2025 |
| Gate | N03 |
| Airline | Cambridge Airways |

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**(b)** The data type for `Gate` is changed to an enumerated data type, `GateID`.

**(i)** Write a **pseudocode** statement to declare `GateID` to hold the identity codes for the airport gates:

N01, N02, N03, W01, W02, W03, W04

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..... [2]

**(ii)** Write the new **pseudocode** statement required to replace the declaration of `Gate` in `Departure`.

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8 Numbers are stored in a computer using binary floating-point representation with:

- 12 bits for the mantissa
- 4 bits for the exponent
- two's complement form for both the mantissa and the exponent.

**(a)** Calculate the normalised binary floating-point representation of +124.4375 in this system. Show your working.

| Mantissa | | | | | | | | | | | | Exponent | | | |
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[3]

**(b)** Calculate the denary value of the following normalised binary floating-point number. Show your working.

| Mantissa | | | | | | | | | | | | Exponent | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |

Working .....

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Denary value .....

[3]

9 **(a)** Random file organisation can be used with a database management system.

State **one** other example of where random file organisation could be used.

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..... [1]

**(b)** Describe how records are organised in random files.

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..... [2]

**(c)** Outline the process that would be used to find a record in a random file.

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10 Numbers are stored in a computer system using binary floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

**(a)** Calculate the normalised binary floating-point representation of –10.8125 in this system.

Show your working.

| mantissa | | | | | | | | | | exponent | | | | | |
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working .....

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[3]

**(b)** Calculate the denary value of the given normalised binary floating-point number.

Show your working.

| mantissa | | | | | | | | | | exponent | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |

working .....

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denary value .....

[3]

11 A programmer is writing a program to manage a video library. They require a user-defined data type.

**(a)** Write **pseudocode** statements to declare the composite data type `VideoLibrary` to hold data about each video in the collection. This data includes:

- identity code (any combination of letters and numbers)
- title
- year released
- date purchased
- format (for example DVD, Blu-ray, 4K, MP4)
- running time (minutes)

Use the most appropriate data type in each case.

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**(b)** Identify **one** field in `VideoLibrary` that could be an efficient enumerated data type and give a reason for your choice.

Field .....

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Reason .....

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[2]

12 Numbers are stored in a computer using binary floating-point representation with:

- 8 bits for the mantissa
- 8 bits for the exponent
- two's complement form for both the mantissa and the exponent.

**(a)** Give the largest normalised positive two's complement binary number that can be stored in this system **and** state its denary equivalent.

The denary answer should be expressed in terms of powers of 2.

**Mantissa** **Exponent**

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Denary .....

[2]

**(b)** Calculate the normalised binary floating-point representation of –3.59375 in this system. Show your working.

**Mantissa** **Exponent**

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Working .....

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**END OF QUESTION PAPER**